INSTITUTO SUPERIOR TÉCNICO

Computação Visual Interactiva

Licenciatura em Engenharia Informática e de Computadores
Alameda

Primeiro Mini Teste

The test has 1h00 duration, tolerance included. **Answer the questions only in the table provided on this page**. Identify the first sheet, and do not separate the sheets from the test! In the test there are blank pages for sketch properly identified. These will not be considered in the evaluation. During the examination, only the pen is allowed. It is also allowed the use of calculator (but it is not needed). **A wrong answer in multiple-choice questions discounts 1/3 of the question quotation.**

NOTE: Write the answer of your options in this table!

Question	Answer
1	
2	
3	
4	
5	
6	
7	
8	
9	

Identificação do Aluno	
Nome:	Número:

1. **[1v]** A definition of an image digital processing can be

A: Given a set of images, produce a world model;
B: The digital image processing from digital devices, for instance a computer;
C: Given a world model, produce a set of images;
D: Physical interpretation of the image pixels;
E: None of the above.

2. **[1v]** In the image digital processing, the middle level processing corresponds to:

A: ... Denoising
B: ... Feature enhancement
C: ... Segmentation and classification
D: ... Scene interpretation
E: None of the above.

3. **[1v]** What is the wave length (approximately) where the visible band images are situated:

A: ... 400 nm a 700 nm
B: ... 400 nm a 2500 nm
C: ... superior a 2500 nm
D: ... 0 nm a 400 nm
E: None of the above.

4. **[1v]** The digital image processing is made of several components**.** One of the components is known as "hardcopy". This component refers to:

A: ... system visualization
B: ... store devices
C: ... specialized image processing hardware
D: ... digitalized systems
E: None of the above

5. **[1v]** In the Human Visual System (HVS) the cones are responsible for what kind of vision ?

A: ... scotopic vision
B: ... mesopic vision
C: ... photopic vision
D: ... focal distance
E: None of the above.

6. **[1v]** Which is the dominant color in the Bayer filter ?

A: ... red
B: ... yellow
C: ... blue
D: ... all the previous to compose the color spectra
E: None of the above.

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7. **[1v]** The concept of sampling corresponds to

A: ... Digitalization of the coordinates
B: ... Digitalization of the amplitudes
C: ... Quantification
D: ... Image amplification
E: None of the above.

8. **[1v]** What is the name of the human eye component responsible for the scotopic vision

A: ... Cones
B: ... Retina
C: ... Iris
D: ... Cornea
E: None of the above.

9. **[1v]** The *field of view* depends on

A: ... Focal distance
B: ... Sensor dimensions
C: ... Focal distance and sensor dimensions
D: ... Angle lens aperture
E: None of the above.

10. **[2v]** Consider the following Image that contains two different intensities, I1 and I2, (the background contains "zero" intensity pixels which aare the empty cells):

					20	**20**	**20**	**20**				
10	**10**	**10**	**10**		**20**	**20**	**20**	**20**	**20**		**20**	
10	**10**	**10**	**10**		**20**	**20**	**20**	**20**	**20**	**20**		
10	**10**	**10**	**10**			**20**	**20**	**20**	**20**	**20**	**20**	
10	**10**	**10**	**10**			**20**	**20**		**20**	**20**	**20**	**20**
10	**10**	**10**										
10	**10**	**10**										

a) Compute the probability of each of the intensities in the image (you can write just the expressions)

Answer I1: _____ Answer I2: _____

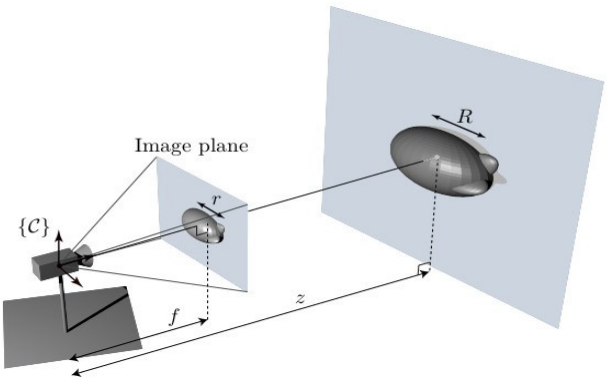
b) What would be the thresholds to separate the two intensities in the image

Answer THS1: _____ Answer THS2: _____

11. **[2v]** To form the digital image two components are necessary, which are

Answer: _____ _____

12. **[2v]** Consider the following figure. Assume that the object has a dimension of 8cm in the image plane. If the focal distance is 4cm, and assuming a distance of 2 meters from the camera to the projected plan. Compute the dimension of the object in that plan.



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13. **[2v]** Given the following binary image,

			1	**1**		
		1	**1**			
		1	**1**	**1**		
					1	

perform the dilation operation with the following structuring element

	1	
1	1	1
	1	

Answer:

-	-	-	-	-	-	-
-						-
-						-
-						-
-						-
-						-
-	-	-	-	-	-	-

14. **[3v]** Consider the following monochromatic image (with very low intensities)

0	**0**	**0**	**0**	**0**
7	**7**	**8**	**8**	**8**
0	**6**	**2**	**2**	**4**
0	**6**	**2**	**2**	**4**
0	**4**	**4**	**4**	**4**

a) Compute the result by applying the correlation of the image with the following mask

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix};$$

-	-	-	-	-
-				-
-				-
-				-
-	-	-	-	-

b) What would be the result with the normalized mean filter ?

-	-	-	-	-
-				-
-				-
-				-
-	-	-	-	-

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